

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		51701	
Total Threats Mitigated	41	Total Packets Analyzed		3155629477	
High Priority	12	Malicious Packets		3149854019	
Low Priority	29	True Negatives Passed		34666	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		15	
False Positives	21	UDP Flood		9	
FP Rate	0.04%	Uncertain		11	
IF Precision	99.91%				
IF Recall	94.19%	Action		Count	
IF F1-Score	96.97%	Quarantined		11	
IF Accuracy	98.00%	Time Ban		10	
		Blackhole		0	
		Released		4	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.91%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.19%	Share of actual attacks successfully flagged
F1-Score	96.97%	Balanced measure combining Precision and Recall
Accuracy	98.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.04%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.81%	Actual attacks that went undetected
True Positive Rate (TPR)	94.19%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.96%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 16239	FN 1001
Actual: Normal	FP 15	TN 33665

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	9135	0	0
Act: ICMP	0	2442	0
Act: UDP	0	0	4662

2c. Response Latency

Metric	Value	Description
Detection Time	429.1 ms	Average time taken to detect an attack
Mitigation Response Time	5.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	2.14%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	49.10%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	16.62%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.6	5	5	2026-09-15 09:42:12	2026-09-15 09:46:07
10.0.0.17	3	3	2026-09-15 09:41:56	2026-09-15 09:45:06
10.0.0.16	2	2	2026-09-15 09:45:12	2026-09-15 09:46:23
10.0.0.18	2	2	2026-09-15 09:45:13	2026-09-15 09:46:24
10.0.0.19	2	2	2026-09-15 09:45:18	2026-09-15 09:46:29
10.0.0.21	1	1	2026-09-15 09:45:44	2026-09-15 09:45:54
10.0.0.22	1	1	2026-09-15 09:45:45	2026-09-15 09:45:55
10.0.0.24	1	1	2026-09-15 09:46:13	2026-09-15 09:46:24
10.0.0.25	1	1	2026-09-15 09:46:14	2026-09-15 09:46:25

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 09:33:09	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:10	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:16	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:19	10.0.0.16	DDoS	SYN Flood	93.2%	Critical	Time Ban (1m)
2026-09-15 09:33:20	10.0.0.16	DDoS	SYN Flood	93.2%	Critical	Time Ban
2026-09-15 09:33:20	10.0.0.19	DDoS	SYN Flood	89.5%	Critical	Quarantined
2026-09-15 09:33:21	10.0.0.18	DDoS	SYN Flood	91.2%	Critical	Time Ban (1m)
2026-09-15 09:33:22	10.0.0.18	DDoS	SYN Flood	91.2%	Critical	Time Ban
2026-09-15 09:33:27	10.0.0.17	DDoS	SYN Flood	93.0%	Critical	Time Ban (1m)
2026-09-15 09:33:27	10.0.0.17	DDoS	SYN Flood	93.0%	Critical	Time Ban
2026-09-15 09:33:31	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-15 09:33:31	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-15 09:33:40	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:40	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:43	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:33:50	10.0.0.20	DDoS	UDP Flood	79.1%	High	Time Ban (1m)
2026-09-15 09:33:50	10.0.0.20	DDoS	UDP Flood	79.1%	High	Time Ban
2026-09-15 09:33:51	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-15 09:33:51	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-15 09:33:53	10.0.0.22	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-15 09:33:54	10.0.0.22	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-15 09:34:10	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:34:10	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:34:12	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 09:34:20	10.0.0.16	DDoS	SYN Flood	93.2%	Critical	Time Ban (2m)
2026-09-15 09:34:20	10.0.0.24	DDoS	ICMP Flood	90.9%	Critical	Time Ban (1m)
2026-09-15 09:34:20	10.0.0.23	DDoS	ICMP Flood	90.7%	Critical	Time Ban (1m)
2026-09-15 09:34:21	10.0.0.23	DDoS	ICMP Flood	90.7%	High	Time Ban
2026-09-15 09:34:21	10.0.0.24	DDoS	ICMP Flood	90.9%	Critical	Time Ban
2026-09-15 09:34:21	10.0.0.18	DDoS	SYN Flood	91.2%	High	Time Ban (2m)
2026-09-15 09:34:22	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban (1m)
2026-09-15 09:34:23	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-15 09:34:28	10.0.0.17	DDoS	SYN Flood	93.0%	Critical	Time Ban (2m)
2026-09-15 09:34:32	10.0.0.19	DDoS	SYN Flood	91.6%	Critical	Time Ban (2m)
2026-09-15 09:36:05	10.0.0.17	DDoS	SYN Flood	92.7%	High	Released
2026-09-15 09:40:00	10.0.0.6	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-15 09:40:11	10.0.0.8	DDoS	Uncertain	68.4%	Low	Released

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 09:40:12	10.0.0.1	DDoS	Uncertain	60.1%	Low	Released
2026-09-15 09:43:13	10.0.0.6	DDoS	UDP Flood	91.0%	High	Released
2026-09-15 09:43:39	10.0.0.6	DDoS	UDP Flood	91.0%	High	Quarantined
2026-09-15 09:45:06	10.0.0.17	DDoS	SYN Flood	94.1%	Critical	Time Ban (5m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6440	92.6%	Critical	Phase 2	2	1m 11s	2026-09-15 09:46:29	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6454	91.2%	Critical	Phase 2	1	11s	2026-09-15 09:46:25	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6388	93.2%	Critical	Phase 2	2	1m 11s	2026-09-15 09:46:24	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6448	91.2%	High	Phase 2	1	11s	2026-09-15 09:46:24	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6279	92.9%	Critical	Phase 2	2	1m 11s	2026-09-15 09:46:23	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6154	91.0%	Critical	Phase 1	5	11s	2026-09-15 09:46:07	Attack Stopped
10.0.0.22	UDP Flood	0.6363	77.4%	High	Phase 2	1	10s	2026-09-15 09:45:55	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6465	82.8%	High	Phase 2	1	10s	2026-09-15 09:45:54	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6516	92.6%	Critical	Phase 2	1	10s	2026-09-15 09:45:28	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6116	91.0%	Critical	Phase 1	4	10s	2026-09-15 09:45:26	Attack Stopped
10.0.0.18	SYN Flood	0.6460	93.2%	Critical	Phase 2	1	10s	2026-09-15 09:45:23	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6383	92.9%	Critical	Phase 2	1	10s	2026-09-15 09:45:22	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6246	94.1%	Critical	Phase 2	3	3m 10s	2026-09-15 09:45:06	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6140	91.0%	High	Phase 1	3	10s	2026-09-15 09:44:35	Attack Stopped
10.0.0.6	UDP Flood	0.6227	91.0%	High	Phase 1	2	10s	2026-09-15 09:43:49	Attack Stopped
10.0.0.6	UDP Flood	0.5736	91.0%	High	Phase 2	1	1m 1s	2026-09-15 09:43:13	Ban Expired

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6246	94.1%	Critical	Phase 2	2	1m 10s	2026-09-15 09:43:06	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6246	94.1%	Critical	Phase 2	1	10s	2026-09-15 09:42:06	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE